

What is a Category?

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Learning outcomes

- Definition
- Representations
- Examples

A category $\mathcal{C}$

It is defined by:

- a set $\text{obj}(\mathcal{C})$ of **objects**
- a set $\mathcal{C}(X, Y)$ of **morphisms** from X to Y , for each $X, Y \in \text{obj}(\mathcal{C})$
- a function assigning to each $Y \in \text{obj}(\mathcal{C})$, the **identity morphism** $\text{id}_Y \in \mathcal{C}(Y, Y)$ of Y
- a function assigning to each $f \in \mathcal{C}(X, Y)$ and $g \in \mathcal{C}(Y, Z)$ (with $X, Y, Z \in \text{obj}(\mathcal{C})$) the **composition of morphisms** $g \circ f \in \mathcal{C}(X, Z)$ satisfying the properties:

① **Unity.**

$\forall f$. if $f \in \mathcal{C}(X, Y)$ then $\text{id}_Y \circ f = f$ and $f \circ \text{id}_X = f$

② **Associativity.**

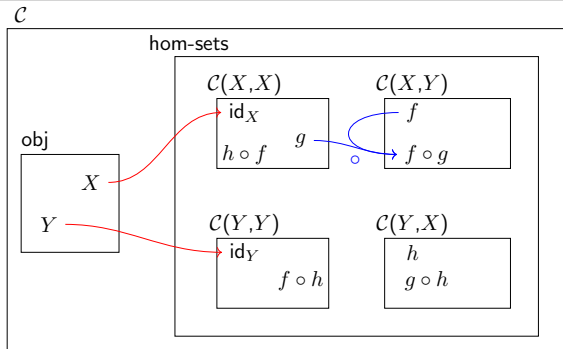
$\forall f, g, h$.

if $f \in \mathcal{C}(X, Y), g \in \mathcal{C}(Y, Z), h \in \mathcal{C}(Z, W)$

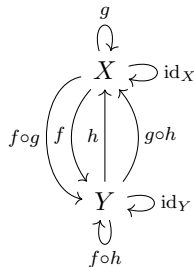
then $h \circ (g \circ f) = (h \circ g) \circ f$

"Pedantic" vs. Essential visual form

"Pedantic"



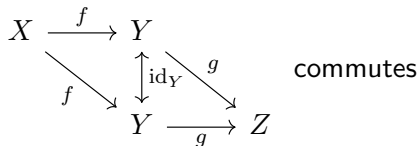
Essential



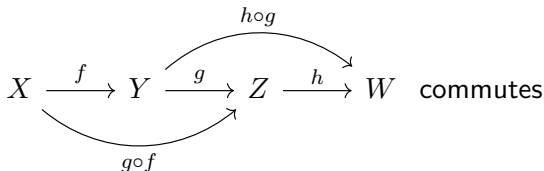
Unity and Associativity as commuting diagrams

For all $f \in \mathcal{C}(X, Y)$, $g \in \mathcal{C}(Y, Z)$, and $h \in \mathcal{C}(Z, W)$,

① $\text{id}_Y \circ f = f$ and $g \circ \text{id}_Y = g$ is equivalent to:



② $h \circ (g \circ f) = (h \circ g) \circ f$ is equivalent to:



Basic examples

Examples of categories

The category **Empty** (**0**).

- Has no objects, i.e. $\text{obj}(\mathbf{0}) = \emptyset$
- Has no morphisms, i.e. it has no hom-sets: $\mathbf{0}(???, ???) = ???$

The category **Terminal** (**1**).

- has a single object, i.e. $\text{obj}(\mathbf{1}) = \{\bullet\}$
- has a single morphism, i.e. $\mathbf{1}(\bullet, \bullet) = \{\text{id}_\bullet\}$ $\bullet \curvearrowright \text{id}_\bullet$
- $\text{id}_\bullet \circ \text{id}_\bullet = \text{id}_\bullet$

Examples of categories

The category **Arrow** ($'0 \rightarrow 1'$).

- has a two objects: $\text{obj}(0 \rightarrow 1) = \{0, 1\}$
- has three morphisms:

$$'0 \rightarrow 1'(0, 0) = \{\text{id}_0\}$$

$$'0 \rightarrow 1'(1, 1) = \{\text{id}_1\}$$

$$'0 \rightarrow 1'(0, 1) = \{f\}$$

- $f \circ \text{id}_0 = f$ and $\text{id}_1 \circ f = f$

$$\begin{array}{ccc} 0 & \xrightarrow{f} & 1 \\ \uparrow \text{id}_0 & & \uparrow \text{id}_1 \end{array}$$

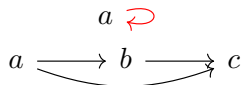
Categories and “orders”

Categories and “orders”

- ① Are the following constructions:
 - ① a pre-order $(X, \leq)$
($\leq$ reflexive, transitive)
 - ② a partial-order $(X, \leq)$
($\leq$ reflexive, transitive, anti-symmetric: $\forall a, b. a \leq b \wedge b \leq a \Rightarrow a = b$)
 - ③ a total-order $(X, \leq)$
($\leq$ reflexive, transitive, anti-symmetric, $\forall a, b. a \leq b \vee b \leq a$)categories?
- ② Are there structural properties that “characterize” pre/partial/total-orders as categories?
- ③ Is an oriented graph $G = (V, E)$ a category?

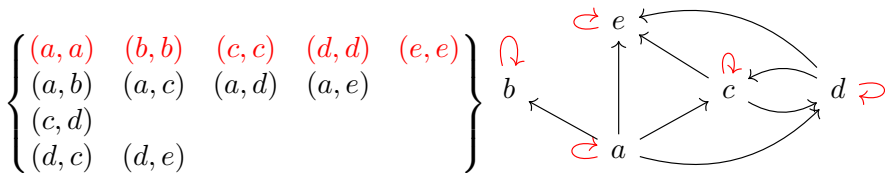
A pre-order $(X, \leq)$ is a category

- $\forall a. a \leq a$
- $\forall a, b, c. a \leq b \wedge b \leq c \Rightarrow a \leq c$



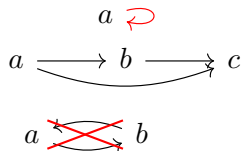
Example.

Let $X = \{a, b, c, d, e\}$. A pre-order on the left, a category on the right:



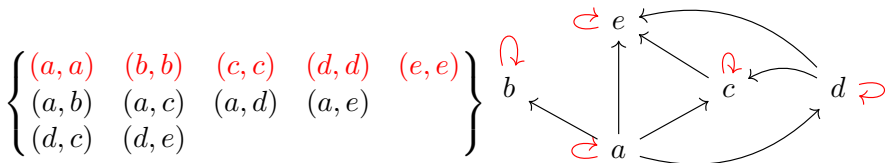
A partial-order is a category

- $\forall a. a \leq a$
- $\forall a, b, c. a \leq b \wedge b \leq c \Rightarrow a \leq c$
- $\forall a, b. a \leq b \wedge b \leq a \Rightarrow a = b$



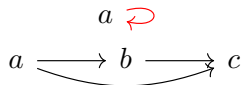
Example 1

Let $X = \{a, b, c, d, e\}$. A partial-order on the left, a category on the right:



A total-order is a category

- $\forall a. a \leq a$
- $\forall a, b, c. a \leq b \wedge b \leq c \Rightarrow a \leq c$
- $\forall a, b. a \leq b \wedge b \leq a \Rightarrow a = b$
- $\forall a, b. a \leq b \vee b \leq a$

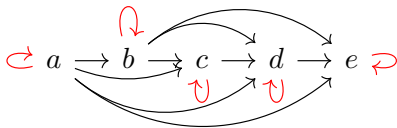


$a \rightarrow b$ or $a \leftarrow b$

Example.

Let $X = \{a, b, c, d, e\}$. A total-order on the left, a category on the right:

$$\left\{ \begin{array}{ccccc} (a, a) & (b, b) & (c, c) & (d, d) & (e, e) \\ (a, b) & (b, c) & (c, d) & (d, e) & \end{array} \right\}$$



Structural properties of orders as categories

Property.

Pre/partial-order have at most one arrow between two objects.

Proof.

 should come from the **non** set $\{(a, a), (b, b), (a, b), (a, b)\}$.



Property.

Total-order have exactly one arrow between two objects.

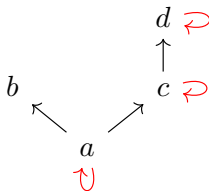
Proof.

 would come from the **non** total order $\{(a, a), (b, b), (a, b), (a, c)\}$.



Not every oriented graph $G = (V, E)$ is a category

Possible reasons in:



- one identity arrow is missing
- the unique arrow required to represent a composition is missing



Milewski, B. and I. Tabachnik (2019). *Category Theory for Programmers*.

<https://bartoszmilewski.com/2014/10/28/category-theory-for-programmers-the-preface/>. Lightning Source UK. ISBN: 9780464243878. URL:

<https://books.google.it/books?id=hA55yQEACAAJ>.